Reviewer 1

The authors analysed how COVID-19 changed mortality patterns in Switzerland. Cardiovascular and mental disorder mortality exccess were correlated to COVID-19 and respiratory disease mortality.

The main limitations are related to missclassification of deaths, model assumptions that oversimplify the results and can bias them. Strenghts rely on the quality of death information system in the country.

Response: We thank the reviewer for these comments regarding strengths and limitations. We agree that potential misclassification of causes of death is a limitation of this type of analysis and this issue has been discussed extensively in the Discussion section (lines 307-318):

"The study also has limitations. First, most analyses rely solely on the unique underlying cause of death, which may overlook mechanisms such as mortality displacement or competing risks. Switzerland's death certificates record immediate, underlying, and concomitant causes, and future work should leverage this richer information to move beyond the underlying cause of death. Incorporating secondary and contributing causes would allow a more rigorous quantification of mortality displacement across time and causes, and would help disentangle misclassification from true shifts in cause-specific mortality. This was illustrated in our preliminary analysis focusing on cancer and more comprehensively in Duràn et al. [17], and could be further extended using more formal modelling approaches. Frailty-survival models capture temporal displacement by accounting for unobserved heterogeneity, while competing-risk models capture cross-cause displacement by redistributing deaths among alternatives [5], [33]. Misclassification, however, may persist even with richer cause-of-death information."

The manuscript "COVID-19 disrupted patterns of cause-specific mortality in Switzerland" addressed an important research question, seeking to understand changes in the patterns of mortality observed before the pandemic. Globally, it is well known that the COVID-19 pandemic changed the mortality patterns. Also, a similar paper was published elsewhere (https://doi.org/10.1016/j.annepidem.2025.12.002). One novelty is understanding cross-cause correlations, but not sure if this is a sufficient novelty to publish the paper.

Comments below:
1) How is this paper different from the other already published papers that were not cited? There are published papers focusing on Switzerland, but the authors did not cite them.

Response: We thank the reviewer for raising the question of novelty, which has led us to revise the introduction. We agree that our contribution needed to be positioned more clearly relative to the existing literature, including the recently published study by Durán et al. (1) and the FSO report on excess mortality by cause (2). Both address related questions but differ from our manuscript in objectives and methodology. The FSO report provides descriptive statistics on changes in the distribution of underlying causes of deaths over years, without counterfactual modelling. Durán et al. provide monthly estimates by broad cause groups and compares methods based on a unique underlying cause of death with methods using weighted multiple causes of deaths, providing evidence that statistics based on the underlying cause of death underestimated chronic disease excess mortality during the pandemic in older adults. Our study differs by modelling multiple causes jointly at a finer level of disaggregation, using a multivariate framework that explicitly accounts for cross-cause correlations and temporal patterns. This allows identification of cause-specific dependencies and patterns. We view our approach as complementary to the existing literature, and believe the methodological contribution extends beyond the Swiss context. Durán et al. focuses on dependencies reflected in the co-occurrence of causes within death certificates, while our analysis focuses on interdependencies in how cause-specific mortality evolves jointly over time after adjusting for seasonality and other temporal patterns.

We adapted the text as follows (Introduction, lines 52-64):

"However, most studies focused on a limited set of causes and on the exceptional circumstances of the pandemic, without capturing the broader interdependence among causes of death. Yet cross-cause correlation is precisely what is needed to disentangle misclassification, mortality displacement, and indirect pathogen effects from one another. In Switzerland, several studies have examined all-cause mortality and described changes in cause-specific mortality during the pandemic [6], [14], [15], and the Federal Statistical Office has documented shifts in the relative distributions of deaths across cause categories over 2020-2022 [16]. Most recently, Durán et al. found lower estimates of excess mortality from chronic diseases when using the underlying rather than multiple causes of death during the COVID-19 wave, supporting the view that COVID-19 may in some cases have displaced chronic conditions as the underlying cause reported on death certificates [17]. Across all these studies, however, mortality series have been analysed separately by cause, and no existing work has jointly modelled their temporal dynamics throughout the pandemic."

2) Although the authors claimed that cross-cause correlation is something not explored yet, it is not sufficiently clear that this is important. Discussion barely approaches this.

Response: We thank the reviewer for this comment. Cross-cause correlations are important because they help distinguish between competing explanations for observed mortality patterns, particularly misclassification, mortality displacement, and indirect effects. We have revised the Introduction and the Discussion to better clarify this rationale:

Introduction (lines 52-56)

"However, most studies focused on a limited set of causes and on the exceptional circumstances of the pandemic, without capturing the broader interdependence among causes of death. Yet cross-cause correlation is precisely what is needed to disentangle misclassification, mortality displacement, and indirect pathogen effects from one another."

Discussion (lines 279-282)

"Modelling these correlations jointly made it possible to distinguish between competing explanations for cause-specific mortality patterns (i.e., misclassification, mortality displacement, and indirect effects) which cannot be disentangled when causes are analysed in isolation."

3) It is not clear why separate diseases in chapter XX are divided into two groups (external and suicide).

Response: We thank the reviewer for this comment. We separated suicide from other external causes to allow a specific assessment of potential changes in suicide mortality during the pandemic, given early concerns that restrictions could affect suicide risk. We have added a sentence in the Methods to clarify this:

Methods (lines 90-92)
"*We modelled suicide separately to assess potential pandemic-related changes, given early concerns regarding the impact of restrictions on suicide mortality.*"

4) Figures should be clearer. For example, I spent considerable time trying to understand Figures 2 and 3. Maybe including the months in the X-axis? I presumed this is a month-by-month analysis. Also, some axes are quite hard to understand without the text (Figure 5, for example). Figure 4 is also hard to understand because it lacks any order or logical sequence.

Response: Figures 2, 3 and 4 are based on weekly aggregated mortality data, as described in the Methods section, and therefore represent weekly estimates of excess mortality. To improve clarity, we have revised the x-axis labels in Figures 3B and 4 (e.g., "2020-01" now shown as "Jan 2020") to make the time scale more immediately interpretable. For Figure 5, additional details have been added in the caption to clarify the meaning of the lag on the x-axis:

Caption of Figure 5

"Partial correlation between excess mortality due to either COVID-19 (red) or to other respiratory diseases (blue, tagged as "Respiratory Diseases") and excess mortality due to cardiovascular diseases, cancers or mental/neurological diseases, for 65-79 and 80+ age groups, considering different lags (in weeks), i.e. different time shifts between changes in excess mortality due to respiratory diseases and changes in excess mortality due to the considered cause."

For Figure 4, the results are presented in four panels (A–D), each corresponding to an age group, and display only the main causes for clarity. The number of causes displayed varies by age group due to differences in mortality counts. Full results for all age groups and causes are available in the Supplementary Material.

5) As the authors used Bayesian Models, it would be good to mention what "CrI" (credible interval) stands for to reach a broader audience.

Response: The term "CrI" (credible interval) is now defined in all figure captions where it appears to improve clarity for a broader audience.

Reviewer 2

Thank you for the opportunity to review the manuscript "COVID-19 disrupted patterns of cause-specific mortality in Switzerland".
The study modelled cause-specific mortality trends in Switzerland from 2011 to 2019 to assess excess mortality during the COVID-19 pandemic in 2020-2021. The results showed several disruptions in mortality patterns compared with prepandemic baseline. Cardiovascular deaths peaked during the autumn 2020 wave, but no cause showed sustained excess across 2020-2021. In individuals aged ≥80 years, non-COVID respiratory and mental/neurological deaths declined, largely offset by COVID-19 deaths, leaving a modest all-cause excess. Cardiovascular and mental/neurological excesses correlated with COVID-19 and other respiratory excesses. The authors concluded that these patterns likely reflect three overlapping mechanisms: reduced circulation of non-COVID-19 respiratory pathogens, unrecognized COVID-19 deaths (especially cardiovascular), and mortality displacement, pointing to possible underestimation of the true respiratory burden.

This is a well-written manuscript which addresses an important and timely epidemiological question, defined adequate baseline years, accounted for temporal variation, and used a sophisticated Bayesian modelling framework to estimate cause-specific excess mortality.
One inherent risk of bias in such studies is potential misclassification of cause of death. Although Swiss death-certificate data are of high quality, misclassification cannot be ruled out, since analyses were based on a single underlying cause of death. However, this limitation was thoroughly discussed by the authors.

Response: We thank the reviewer for this positive assessment. We agree that misclassification cannot be fully addressed when analyses rely solely on the underlying cause of death. As discussed in the Discussion section and in response to a comment from Reviewer 1, this limitation highlights the need for models that incorporate the multiple causes reported on death certificates, for example through competing-risk approaches. Developing such models would be an important direction for future work, although it raises additional methodological challenges. The following revised passage in the Discussion (lines 307-318) mentioned this concern:

"The study also has limitations. First, most analyses rely solely on the unique underlying cause of death, which may overlook mechanisms such as mortality displacement or competing risks. Switzerland's death certificates record immediate, underlying, and concomitant causes, and future work should leverage this richer information to move beyond the underlying cause of death. Incorporating secondary and contributing causes would allow a more rigorous quantification of mortality displacement across time and causes, and would help disentangle misclassification from true shifts in cause-specific mortality. This was illustrated in our preliminary analysis focusing on cancer and more comprehensively in Duràn et al. [17], and could be further extended using more formal modelling approaches. Frailty-survival models capture temporal displacement by accounting for unobserved heterogeneity, while competing-risk models capture cross-cause displacement by redistributing deaths among alternatives [5], [33]. Misclassification, however, may persist even with richer cause-of-death information."

At a general methodological level, the analytical strategy appears thoughtful and coherent, particularly in its treatment of seasonality, long-term temporal trends, and the correlation across causes of death. The results section is well structured, progressing from descriptive mortality patterns to pre-pandemic baseline trends and then to pandemic-period excess mortality. The interpretation remains appropriately grounded in the results and does not overreach the data, and the four proposed broad disruptions to explain the observed patterns are plausible.

Major comment:
The kind of statistical modelling in this manuscript is outside my area of methodological expertise and I recommend that the manuscript should also be evaluated by a reviewer with this specific kind of expertise.
Minor comments:
Sometimes inconsistent use of the present and past tenses in the manuscript, please check and adjust accordingly.

Response: We thank the reviewer for this observation. We have revised the manuscript to ensure consistent use of verb tenses throughout.

References

1. Durán N D, Kaufman JS, Chiolero A, Carmeli C. Multiple causes of death data to track trends of mortality from chronic diseases: Insights from the COVID-19 pandemic in Switzerland. Ann Epidemiol. 1 janv 2026;113:23-9. doi:10.1016/j.annepidem.2025.12.002

2. {Federal Statistical Office}. The impact of the COVID-19 pandemic on -mortality and causes of death in Switzerland [Internet]. [cité 18 mai 2026]. Disponible sur: https://www.swissstats.bfs.admin.ch/data/webviewer/appId/ch.admin.bfs.swissstat/article/issue231412602200-01/package